

### Fluid, Electrolyte and Acid-Base Balance

#### -Fluid Imbalances:-

- Dehydration.
- Hypovolemia.
- Hypervolemia.
- Water intoxication.

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#### **1. Dehydration:-**

- Loss of body fluids → increased concentration of solutes in the blood and a rise in serum Na<sup>+</sup> levels.
- Fluid shifts out of cells into the blood to restore balance.
- Cells shrink from fluid loss and can no longer function properly.

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#### **2. Hypovolemia:-**

- Isotonic fluid loss from the extracellular space.
- Can progress to hypovolemic shock.
- Caused by:
  - Excessive fluid loss (hemorrhage).
  - Decreased fluid intake.
  - Third space fluid shifting.

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#### **3. Hypervolemia:-**

- Excess fluid in the extracellular compartment as a result of fluid or sodium retention, excessive intake, or renal failure.
- Occurs when compensatory mechanisms fail to restore fluid balance.
- Leads to CHF and pulmonary edema.

#### **❖ Edema:**

- Fluid is forced into tissues by the hydrostatic pressure.
- First seen in dependent areas.
- Anasarca - severe generalized edema.
- Pitting edema.
- Pulmonary edema.

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#### **4. Water Intoxication:-**

- Hypotonic extracellular fluid shifts into cells to attempt to restore balance.
- Cells swell.
- Causes:-
  - SIADH. –Rapid infusion of hypotonic solution.
  - Excessive tap water NG irrigation or enemas. – Psychogenic polydipsia.

## Electrolytes

- Charged particles in solution.
- Cations (+).
- Anions (-).
- Integral part of metabolic and cellular processes.

### – Positive or Negative?

| <u>Cations (+)</u> | <u>Anions (-)</u> |
|--------------------|-------------------|
| Sodium             | Chloride          |
| Potassium          | Bicarbonate       |
| Calcium            | Phosphate         |
| Magnesium          | Sulfate           |

### – Major Cations:

- EXTRACELLULAR :- Sodium (Na<sup>+</sup>).
- INTRACELLULAR:- POTASSIUM (K<sup>+</sup>).

### -Electrolyte Imbalances:-

- Hyponatremia/ hypernatremia.
- Hypokalemia/ Hyperkalemia.
- Hypomagnesemia/ Hypermagnesemia.
- Hypocalcemia/ Hypercalcemia.
- Hypophosphatemia/ Hyperphosphatemia.
- Hypochloremia/ Hyperchloremia.

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### 1. Sodium:

- Major extracellular cation.
- Attracts fluid and helps preserve fluid volume.
- Combines with chloride and bicarbonate to help regulate acid-base balance.
- Normal range of serum sodium 135 - 145 mEq/L.

#### ❖ Sodium and Water:

- If sodium intake suddenly increases, extracellular fluid concentration also rises.
- Increased serum Na<sup>+</sup> increases thirst and the release of ADH, which triggers kidneys to retain water.
- Aldosterone also has a function in water and sodium conservation when serum Na<sup>+</sup> levels are low.

#### ❖ Sodium-Potassium Pump:

- Sodium (abundant outside cells) tries to get into cells.
- Potassium (abundant inside cells) tries to get out of cells.
- Sodium-potassium pump maintains normal concentrations.

- Pump uses ATP, magnesium and an enzyme to maintain sodium-potassium concentrations.
- Pump prevents cell swelling and creates an electrical charge allowing neuromuscular impulse transmission.

#### ❖ Hyponatremia:

- Serum Na<sup>+</sup> level < 135 mEq/L.
- Deficiency in Na<sup>+</sup> related to amount of body fluid.
- Several types:
  - Dilutional.
  - Depletional.
  - Hypovolemic.
  - Hypervolemic.
  - Isovolemic.

#### ❖ Types of Hyponatremia:

- Dilutional - results from Na<sup>+</sup> loss, water gain.
- Depletional - insufficient Na<sup>+</sup> intake.
- Hypovolemic - Na<sup>+</sup> loss is greater than water loss; can be renal (diuretic use) or non-renal (vomiting).
- Hypervolemic - water gain is greater than Na<sup>+</sup> gain; edema occurs.
- Isovolumic - normal Na<sup>+</sup> level, too much fluid.

#### ❖ Hypernatremia:

- Excess Na<sup>+</sup> relative to body water.
- Occurs less often than hyponatremia.
- Thirst is the body's main defense.
- When hypernatremia occurs, fluid shifts outside the cells.
- May be caused by water deficit or over-ingestion of Na<sup>+</sup>.
- Also may result from diabetes insipidus.

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## 2. Potassium:

- Major intracellular cation.
- Untreated changes in K<sup>+</sup> levels can lead to serious neuromuscular and cardiac problems.
- Normal K<sup>+</sup> levels = 3.5 - 5 mEq/L.

#### ❖ Balancing Potassium:

- Most K<sup>+</sup> ingested is excreted by the kidneys
- Three other influential factors in K<sup>+</sup> balance :
  1. Na<sup>+</sup>/K<sup>+</sup> pump.
  2. Renal regulation.
  3. pH level.

❖ **Sodium/Potassium Pump:**

- Uses ATP to pump potassium into cells.
- Pumps sodium out of cells.
- Creates a balance.

❖ **Renal Regulation:**

- Increased K<sup>+</sup> levels → increased K<sup>+</sup> loss in urine.
- Aldosterone secretion causes Na<sup>+</sup> reabsorption and K<sup>+</sup> excretion.

❖ **pH:**

- Potassium ions and hydrogen ions exchange freely across cell membranes.
- Acidosis → hyperkalemia (K<sup>+</sup> moves out of cells).
- Alkalosis → hypokalemia (K<sup>+</sup> moves into cells).

❖ **Hypokalemia:**

- Serum K<sup>+</sup> < 3.5 mEq/L.
- Can be caused by GI losses, diarrhea, insufficient intake, non-K<sup>+</sup> sparing diuretics (thiazide, furosemide).

❖ **Hyperkalemia:**

- Serum K<sup>+</sup> > 5 mEq/L.
- Less common than hypokalemia.
- Caused by altered kidney function, increased intake (salt substitutes), blood transfusions, meds (K<sup>+</sup>-sparing diuretics), cell death (trauma).

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**3. Magnesium:**

- Helps produce ATP.
- Role in protein synthesis & carbohydrate metabolism.
- Helps cardiovascular system function (vasodilation).
- Regulates muscle contractions.

❖ **Hypomagnesemia:**

- Serum Mg<sup>++</sup> level < 1.5 mEq/L
- Caused by poor dietary intake, poor GI absorption, excessive GI/urinary losses
- High risk clients:
  - Chronic alcoholism
  - Malabsorption
  - GI/urinary system disorders
  - Sepsis
  - Burns
  - Wounds needing debridement

❖ **Hypermagnesemia:**

- Serum Mg<sup>++</sup> level > 2.5 mEq/L.
- Not common.

- Renal dysfunction is most common cause:
    - Renal failure.
    - Addison's disease.
    - Adrenocortical insufficiency.
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#### 4. Calcium:

- 99% in bones, 1% in serum and soft tissue (measured by serum  $\text{Ca}^{++}$ ).
- Works with phosphorus to form bones and teeth.
- Role in cell membrane permeability.
- Affects cardiac muscle contraction.
- Participates in blood clotting.

#### ❖ Calcium Regulation:

- Affected by body stores of  $\text{Ca}^{++}$  and by dietary intake & Vitamin D intake.
- Parathyroid hormone draws  $\text{Ca}^{++}$  from bones increasing low serum levels (Parathyroid pulls).
- With high  $\text{Ca}^{++}$  levels, calcitonin is released by the thyroid to inhibit calcium loss from bone (Calcitonin keeps).

#### ❖ Hypocalcemia:

- Serum calcium < 8.9 mg/dl.
- Ionized calcium level < 4.5 mg/Dl.
- Caused by inadequate intake, malabsorption, pancreatitis, thyroid or parathyroid surgery, loop diuretics, low magnesium levels.

#### ❖ Hypercalcemia:

- Serum calcium > 10.1 mg/dl.
  - Ionized calcium > 5.1 mg/dl.
  - Two major causes:
    - Cancer.
    - Hyperparathyroidism.
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#### 5. Phosphorus:

- The primary anion in the intracellular fluid.
- Crucial to cell membrane integrity, muscle function, neurologic function and metabolism of carbs, fats and protein.
- Functions in ATP formation, phagocytosis, platelet function and formation of bones and teeth.

#### ❖ Hypophosphatemia:

- Serum phosphorus < 2.5 mg/dl.
- Can lead to organ system failure.
- Caused by respiratory alkalosis (hyperventilation), insulin release, malabsorption, diuretics, elevated parathyroid hormone levels, extensive burns.

❖ **Hyperphosphatemia:**

- Serum phosphorus > 4.5 mg/dl.
  - Caused by impaired kidney function, cell damage, hypoparathyroidism, respiratory acidosis, increased dietary intake.
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**6. Chloride:**

- Major extracellular anion.
- Sodium and chloride maintain water balance.
- Secreted in the stomach as hydrochloric acid.
- Aids carbon dioxide transport in blood.

❖ **Hypochloremia:**

- Serum chloride < 96 mEq/L.
- Caused by decreased intake or decreased absorption, metabolic alkalosis, and loop, osmotic or thiazide diuretics.

❖ **Hyperchloremia:**

- Serum chloride > 106 mEq/L.
  - Rarely occurs alone.
  - Caused by dehydration, renal failure, respiratory alkalosis, salicylate toxicity, hyperpara-thyroidism, hyperaldosteronism, hyponatremia.
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**Chemical Buffers:-**

- Immediate acting.
  - Combine with offending acid or base to neutralize harmful effects until another system takes over.
  - Bicarb buffer - mainly responsible for buffering blood and interstitial fluid.
  - Phosphate buffer - effective in renal tubules.
  - Protein buffers - most plentiful – hemoglobin.
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**Respiratory System:-**

- Lungs regulate blood levels of CO<sub>2</sub>.
  - CO<sub>2</sub> + H<sub>2</sub>O = Carbonic acid.
  - High CO<sub>2</sub> = slower breathing (hold on to carbonic acid and lower pH).
  - Low CO<sub>2</sub> = faster breathing (blow off carbonic acid and raise pH).
  - Twice as effective as chemical buffers, but effects are temporary.
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**Kidneys:-**

- Reabsorb or excrete excess acids or bases into urine.
- Produce bicarbonate.
- Adjustments by the kidneys take hours to days to accomplish.
- Bicarbonate levels and pH levels increase or decrease together.

## **Arterial Blood Gases (ABG):-**

- Uses blood from an arterial puncture.
  - Three test results relate to acid-base balance:
    - pH.
    - PaCO<sub>2</sub>.
    - HCO<sub>3</sub>.
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## **Acid-Base Imbalances:-**

### **1. Respiratory Acidosis:**

- Any compromise in breathing can result in respiratory acidosis.
- Hypoventilation → carbon dioxide buildup and drop in pH.
- Can result from neuromuscular trouble, depression of the brain's respiratory center, lung disease or airway obstruction.

### **2. Respiratory Alkalosis:**

- Most commonly results from hyperventilation caused by pain, salicylate poisoning, use of nicotine or aminophylline, hypermetabolic states or acute hypoxia (over stimulates the respiratory center).

### **3. Metabolic Acidosis:**

- Characterized by gain of acid or loss of bicarb.
- Associated with ketone bodies:  
Diabetes mellitus, alcoholism, starvation, hyperthyroidism.
- Other causes:  
Lactic acidosis secondary to shock, heart failure, pulmonary disease, hepatic disease, seizures, strenuous exercise.

### **4. Metabolic Alkalosis:**

- Commonly associated with hypokalemia from diuretic use, hypochloremia and hypocalcemia.
- Also caused by excessive vomiting, NG suction, Cushing's disease, kidney disease or drugs containing baking soda.